

# The Market for Inflation Risk

Saleem Bahaj, Robert Czech, Sitong Ding, Ricardo Reis

Presented by Janet Choi

Dec 6, 2023

# Overview

## Motivation:

- Inflation risk is a concern for many investors
- Market for **inflation swaps**: Allows investors to hedge against inflation risk
  - ⇒ Trade based on expectations of future inflation
  - ⇒ Can provide insight into **market's inflation expectations**

► Swap Prices

## This paper:

- Present a model of supply and demand in the inflation swap market
- Use high-frequency data on swap trades to study market responses to different shocks

## Research questions:

- Who are the main traders of inflation risk and what determines their holdings?
- What are the relevant shocks that drive prices in this market?
- Are swap rates reasonable measures of expected inflation?

# Overview

## Motivation:

- Inflation risk is a concern for many investors
- Market for **inflation swaps**: Allows investors to hedge against inflation risk
  - ⇒ Trade based on expectations of future inflation
  - ⇒ Can provide insight into **market's inflation expectations**

► Swap Prices

## This paper:

- Present a model of supply and demand in the inflation swap market
- Use high-frequency data on swap trades to study market responses to different shocks

## Research questions:

- Who are the main traders of inflation risk and what determines their holdings?
- What are the relevant shocks that drive prices in this market?
- Are swap rates reasonable measures of expected inflation?

# Preview of Results

- Data show evidence of **segmentation** in the inflation swap market
  - Short-horizon swap market: pension funds as main traders
  - Long-horizon swap market: hedge funds as main traders
- Model: three sectors (pension funds, hedge funds, dealer banks)
  - Each institution in each sector solves a portfolio allocation problem
- Price/quantity movements for ST inflation protection largely determined by shocks to supply/demand of swaps
- Prices for **LT** swaps mainly driven by **shocks to fundamental expected inflation**
- Present a 'cleaned' measure of LT inflation swap prices → suggests possibility that observed LT swap prices **overstate** market's expectations of inflation

# Outline

1. UK Inflation Swap Market
2. Model
3. Identification
4. Estimates and Applications

# 1. UK Inflation Swap Market

## 2. Model

## 3. Identification

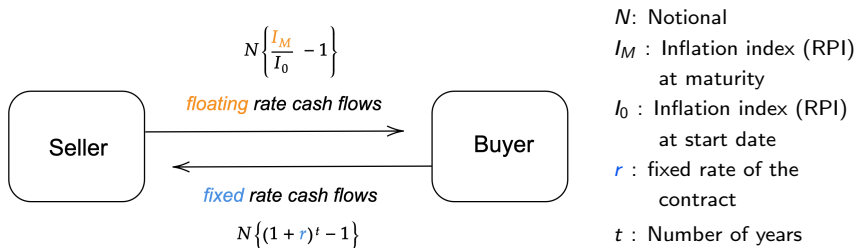
## 4. Estimates and Applications

### 4.1 Impulse Responses

### 4.2 Historical Decomposition

# Inflation Swaps

- Contract between two parties with the purpose of **transferring inflation risk**
  - By exchanging cash flows
  - Mostly 'zero coupon' swaps: exchange payments only at maturity
  - Maturities typically range from 1 year - over 30 years



- Fixed rate reflects expected inflation at maturity
- If  $\Delta$  inflation  $>$  compounded fixed rate, buyer profits from swap

# Data

- Use EMIR Trade Repository (TR) data
  - European Market Infrastructure Regulation (EMIR)
    - UK legal entities must report terms of any derivative transaction to a trade repository authorized by the Financial Conduct Authority by the next business day
  - Trade-level data on OTC inflation swaps
  - All trades submitted to DTCC Derivatives Repository Plc with at least one party being a UK-regulated entity
    - Daily trade state reports
  - Daily frequency, Jan 2nd 2019 - Feb 10th 2023



# Market Size and Main Participants

Market size (as measured by gross notional amount)

- \$3.5 - 4 trillion,  $\approx$  110-130% of UK GDP [▶ Figure](#)

Focus on the dealer-client segment of the UK inflation swap market

- Dealer banks sell inflation swaps to different institutions
  - Pension funds and liability driven investors (PFLDIs)
  - Hedge funds
  - Others (non-dealer banks, insurance companies ...)
- Over-the-counter (OTC) market

# Key Facts on the UK Inflation Swap Market

1. **Dealer banks** are *not* neutral market makers, but rather net sellers of inflation protection ▶ Fact 1
2. **PFLDI**s are buyers in **long** horizon markets (10 yrs or more)
  - Persistently large positive net positions ▶ Fact 2
  - Hedge long-term liabilities (future pension payments)
  - Dealers meet this demand: large negative net positions
3. **Hedge funds** trade mainly in **short** horizon markets (3 yrs or less)
  - Switch between net buyers and sellers ▶ Fact 3
  - Consistent with hedge funds being arbitrageurs

These facts together imply **segmentation** of the market

⇒ Pension funds trade in LT market, hedge funds in ST market,  
dealer banks active in both

⇒ Model: based on these observations (imposed as assumptions)

# Key Facts on the UK Inflation Swap Market

1. **Dealer banks** are *not* neutral market makers, but rather net sellers of inflation protection ▶ Fact 1
2. **PFLDI**s are buyers in **long** horizon markets (10 yrs or more)
  - Persistently large positive net positions ▶ Fact 2
  - Hedge long-term liabilities (future pension payments)
  - Dealers meet this demand: large negative net positions
3. **Hedge funds** trade mainly in **short** horizon markets (3 yrs or less)
  - Switch between net buyers and sellers ▶ Fact 3
  - Consistent with hedge funds being arbitrageurs

These facts together imply **segmentation** of the market

⇒ Pension funds trade in LT market, hedge funds in ST market,  
dealer banks active in both

⇒ Model: based on these observations (imposed as assumptions)

# Key Facts on the UK Inflation Swap Market

1. **Dealer banks** are *not* neutral market makers, but rather net sellers of inflation protection ▶ Fact 1
2. **PFLDI**s are buyers in **long** horizon markets (10 yrs or more)
  - Persistently large positive net positions ▶ Fact 2
  - Hedge long-term liabilities (future pension payments)
  - Dealers meet this demand: large negative net positions
3. **Hedge funds** trade mainly in **short** horizon markets (3 yrs or less)
  - Switch between net buyers and sellers ▶ Fact 3
  - Consistent with hedge funds being arbitrageurs

These facts together imply **segmentation** of the market

⇒ Pension funds trade in LT market, hedge funds in ST market,  
dealer banks active in both

⇒ Model: based on these observations (imposed as assumptions)

# Key Facts on the UK Inflation Swap Market

1. **Dealer banks** are *not* neutral market makers, but rather net sellers of inflation protection ▶ Fact 1
2. **PFLDI**s are buyers in **long** horizon markets (10 yrs or more)
  - Persistently large positive net positions ▶ Fact 2
  - Hedge long-term liabilities (future pension payments)
  - Dealers meet this demand: large negative net positions
3. **Hedge funds** trade mainly in **short** horizon markets (3 yrs or less)
  - Switch between net buyers and sellers ▶ Fact 3
  - Consistent with hedge funds being arbitrageurs

These facts together imply **segmentation** of the market

⇒ Pension funds trade in LT market, hedge funds in ST market,  
dealer banks active in both

⇒ Model: based on these observations (imposed as assumptions)

1. UK Inflation Swap Market

2. Model

3. Identification

4. Estimates and Applications

4.1 Impulse Responses

4.2 Historical Decomposition

# Model

## Three types ('sectors') of agents

- Pension funds ( $f$ ), hedge funds ( $h$ ), dealer banks ( $b$ )
- Many institutions ( $i$ ) within each type
  - $\Theta_x, x \in \{f, h, b\}$ : set of institutions in each sector
- Each type solves a static portfolio allocation problem

## Agents can trade in three types of assets

- Single period = a trading day
- Inflation swap contract for the short (*long*) horizon
  - Pay **fixed rate**  $P$  ( $p$ ), receive **floating rate**  $\Pi$  ( $\pi$ ) at end of day
  - Holdings:  $Q_{x,i}$  ( $q_{x,i}$ ),  $x \in \{f, h, b\}$
  - Subject to capacity constraints
- Market asset: cost  $s$ , stochastic payoff  $d$ , holdings  $e_{x,i}$
- Risk-free asset: gross risk-free return normalized to 1

# Model

## Three types ('sectors') of agents

- Pension funds ( $f$ ), hedge funds ( $h$ ), dealer banks ( $b$ )
- Many institutions ( $i$ ) within each type
  - $\Theta_x, x \in \{f, h, b\}$ : set of institutions in each sector
- Each type solves a static portfolio allocation problem

## Agents can trade in three types of assets

- Single period = a trading day
- Inflation swap contract for the short (*long*) horizon
  - Pay **fixed rate**  $P$  ( $p$ ), receive **floating rate**  $\Pi$  ( $\pi$ ) at end of day
  - Holdings:  $Q_{x,i}$  ( $q_{x,i}$ ),  $x \in \{f, h, b\}$
  - Subject to capacity constraints
- Market asset: cost  $s$ , stochastic payoff  $d$ , holdings  $e_{x,i}$
- Risk-free asset: gross risk-free return normalized to 1



# Model

## Capacity constraint

$$G_x(q_{x,i}, z_{x,i}) \geq 0, \quad x \in \{f, h, b\}$$

- Constraints on each institution's ability to take on inflation risk (regulatory constraints, limitations on short sales, ...)
- $G_x(\cdot)$ : continuous, measures the proximity of the institution to the capacity limit
- $z_{x,i}$ : exogenous institution-specific parameter, determines tightness
- $\lambda_{x,i}$ : Lagrange multiplier associated with the constraint

**Idiosyncratic risk:**  $y_{x,i}$ , mean zero, cannot be traded (incomplete markets)

**Assumption (Segmented markets):** Key Facts

**Pension** funds *do not* participate in the **short**-horizon market ( $Q_{f,i} = 0$ )  
and **hedge** funds *do not* participate in the **long** horizon market ( $q_{h,i} = 0$ )

# Model

## Capacity constraint

$$G_x(q_{x,i}, z_{x,i}) \geq 0, \quad x \in \{f, h, b\}$$

- Constraints on each institution's ability to take on inflation risk (regulatory constraints, limitations on short sales, ...)
- $G_x(\cdot)$ : continuous, measures the proximity of the institution to the capacity limit
- $z_{x,i}$ : exogenous institution-specific parameter, determines tightness
- $\lambda_{x,i}$ : Lagrange multiplier associated with the constraint

**Idiosyncratic risk:**  $y_{x,i}$ , mean zero, cannot be traded (incomplete markets)

**Assumption (Segmented markets):** ◀ Key Facts

**Pension** funds *do not* participate in the **short**-horizon market ( $Q_{f,i} = 0$ )  
and **hedge** funds *do not* participate in the **long** horizon market ( $q_{h,i} = 0$ )

# Model

## Capacity constraint

$$G_x(q_{x,i}, z_{x,i}) \geq 0, \quad x \in \{f, h, b\}$$

- Constraints on each institution's ability to take on inflation risk (regulatory constraints, limitations on short sales, ...)
- $G_x(\cdot)$ : continuous, measures the proximity of the institution to the capacity limit
- $z_{x,i}$ : exogenous institution-specific parameter, determines tightness
- $\lambda_{x,i}$ : Lagrange multiplier associated with the constraint

**Idiosyncratic risk:**  $y_{x,i}$ , mean zero, cannot be traded (incomplete markets)

**Assumption (Segmented markets):** ◀ Key Facts

**Pension** funds *do not* participate in the **short**-horizon market ( $Q_{f,i} = 0$ )  
and **hedge** funds *do not* participate in the **long** horizon market ( $q_{h,i} = 0$ )

# Pension Fund $i$ 's Problem

$$\max_{q_{f,i}, e_{f,i}} \mathbb{E}_{f,i} [-\exp(-\tilde{\gamma}_{f,i} a'_{f,i})] \quad \text{with} \quad \tilde{\gamma}_{f,i} = \gamma_{f,i} / a_{f,i}$$

$$\text{s.t.} \quad a'_{f,i} = a_{f,i} + \underbrace{(\pi - p)q_{f,i}}_{\text{LT inflation swap}} + \underbrace{(d - s)e_{f,i}}_{\text{market asset}} + y_{f,i}$$

$$G_f(Q_{f,i}, z_{f,i}) \geq 0$$

- $a_{f,i}$ : initial position,  $a'_{f,i}$ : terminal market value of  $i$ 's chosen portfolio
- 'Budget constraint', but unclear how resources move
- Pays swap fixed rate  $p$ , receives floating inflation rate  $\pi$ 
  - Each pension fund  $i$  may disagree on expected inflation (captured in  $\mu_{f,i}$ )

$$\mathbb{E}_{f,i}[\pi] = \mu_{f,i} \pi^e \quad \text{with} \quad \sum_{i \in \Theta_f} \mu_{f,i} = 1$$

- $\pi^e$ : **fundamental** expected inflation
- Agree on expectations of market asset payoff, background risk

$$\mathbb{E}_{f,i}[d] = \theta_d, \quad \mathbb{E}_{f,i}[y_{f,i}] = 0$$

# Hedge Fund $i$ 's Problem

$$\begin{aligned} \max_{Q_{h,i}, e_{h,i}} \quad & \mathbb{E}_{h,i} \left[ -\exp \left( -\tilde{\gamma}_{h,i} a'_{h,i} \right) \right] \quad \text{with} \quad \tilde{\gamma}_{h,i} = \gamma_{h,i} / a_{h,i} \\ \text{s.t.} \quad & a'_{h,i} = a_{h,i} + \underbrace{(\Pi - P)Q_{h,i}}_{\text{ST inflation swap}} + \underbrace{(d - s)e_{h,i}}_{\text{market asset}} + y_{h,i} \\ & G_h(Q_{h,i}, z_{h,i}) \geq 0 \end{aligned}$$

- By assumption, only trade in short-horizon swap market
- Beliefs (on expected ST inflation) of individual hedge fund

$$\mathbb{E}_{h,i}[\Pi] = \mu_{h,i} \Pi^e \quad \text{with} \quad \sum_{i \in \Theta_h} \mu_{h,i} = 1$$

## Dealer Bank $i$ 's Problem

$$\begin{aligned} \max_{q_{b,i}, Q_{b,i}, e_{b,i}} \quad & \mathbb{E}_{b,i} \left[ -\exp \left( -\tilde{\gamma}_{b,i} a'_{b,i} \right) \right] \quad \text{with} \quad \tilde{\gamma}_{b,i} = \gamma_{b,i} / a_{b,i} \\ \text{s.t.} \quad & a'_{b,i} = a_{b,i} + \underbrace{(\pi - p)q_{b,i}}_{\text{LH swap}} + \underbrace{(\Pi - P)Q_{b,i}}_{\text{SH swap}} + \underbrace{(d - s)e_{b,i}}_{\text{market asset}} + y_{b,i} \end{aligned}$$

- Beliefs of each dealer bank institution

$$\mathbb{E}_{b,i}[\pi] = \mu_{b,i} \pi^e, \quad \mathbb{E}_{b,i}[\Pi] = \mu_{b,i} \Pi^e \quad \text{with} \quad \sum_{i \in \Theta_f} \mu_{b,i} = 1$$

**Assumption (Desk separation within the day):** Dealers face separate capacity constraints

$$G_b^S(Q_{b,i}, z_{b,i}) \geq 0 \quad \text{and} \quad G_b^L(q_{b,i}, z_{b,i}) \geq 0$$

so that  $\partial G_b^S(\cdot, \cdot) / \partial q_{b,i} = 0$  and  $\partial G_b^L(\cdot, \cdot) / \partial Q_{b,i} = 0$ .

- Risk-taking capacity of ST desk not constrained by risk taken by LT desk

# Demand and Supply

Prop.1 : Given market prices  $p^*$  and  $s^*$ , a **pension fund's optimal demand for inflation protection**  $q_{f,i}^*$  scaled by size is given by

$$\frac{q_{f,i}^*}{a_{f,i}} = \underbrace{\frac{\mu_{f,i}\pi^e - p^*}{\gamma_{f,i}\sigma_\pi^2(1 - \rho_{\pi,d}^2)}}_{\text{subj. expected excess return}} - \underbrace{\left(\frac{\sigma_d}{\sigma_\pi}\right) \left[ \frac{\theta_d - s^*}{\gamma_{f,i}\sigma_d^2(1 - \rho_{\pi,d}^2)} \right]}_{\text{hedging demand}} \rho_{\pi,d} - \underbrace{\left[ \frac{1}{(1 - \rho_{\pi,d}^2)\sigma_\pi^2} \right] \left( \frac{\sigma_{\pi,y_{f,i}}}{a_{f,i}} + \frac{\lambda_{f,i}g_{f,i}}{\gamma_{f,i}} \right)}_{\text{liquidity frictions}}$$

- Downward sloping in price and risk aversion ( $\gamma_{f,i}$ )
- Pension fund  $i$  demands more inflation protection if
  - (1) Expects a higher inflation ( $\uparrow \mu_{f,i}\pi^e$ ), less uncertain about inflation ( $\downarrow \sigma_\pi$ ), less risk averse ( $\downarrow \gamma_{f,i}$ )
  - (2) Lower correlation betw. expected inflation and returns on market assets ( $\downarrow \rho_{\pi,d}$ )
  - (3) Lower *Cov(expected inflation, background income)* ( $\downarrow \sigma_{\pi,y_{f,i}}$ ), Loose/non-binding capacity constraint ( $\downarrow \lambda_{f,i}g_{f,i}$ )

# Demand and Supply

Prop.2 : Given market prices  $P^*$ ,  $p^*$  and  $s^*$ , the **optimal demand** in the two markets of a **dealer bank** is:

$$(LT) \quad \frac{q_{b,i}^*}{a_{b,i}} = \frac{\mu_{b,i}\pi^e - p^*}{\gamma_{b,i}\sigma_\pi^2(1 - \rho_{\pi,d}^2)} - \left(\frac{\sigma_d}{\sigma_\pi}\right) \left[ \frac{\theta_d - s^*}{\gamma_{b,i}\sigma_d^2(1 - \rho_{\pi,d}^2)} \right] \rho_{\pi,d} \\ - \left[ \frac{1}{(1 - \rho_{\pi,d}^2)\sigma_\pi^2} \right] \left( \frac{\sigma_{\pi,y_{b,i}}}{a_{b,i}} + \frac{\lambda_{b,i}g_{b,i}}{\gamma_{b,i}} \right)$$

$$(ST) \quad \frac{Q_{b,i}^*}{a_{b,i}} = \frac{\mu_{b,i}\Pi^e - P^*}{\gamma_{b,i}\sigma_\Pi^2(1 - \rho_{\Pi,d}^2)} - \left(\frac{\sigma_d}{\sigma_\Pi}\right) \left[ \frac{\theta_d - s^*}{\gamma_{b,i}\sigma_d^2(1 - \rho_{\Pi,d}^2)} \right] \rho_{\Pi,d} \\ - \left[ \frac{1}{(1 - \rho_{\Pi,d}^2)\sigma_\Pi^2} \right] \left( \frac{\sigma_{\Pi,y_{b,i}}}{a_{b,i}} + \frac{\lambda_{b,i}g_{b,i}}{\gamma_{b,i}} \right)$$



# Market Clearing

Only stated for market for inflation protection

$$(LT) \quad \sum_{i \in \Theta_f} q_{f,i}^* = - \underbrace{\sum_{i \in \Theta_b} q_{b,i}^*}_{\text{LT supply}} \equiv q^*$$

$$(ST) \quad \sum_{i \in \Theta_h} Q_{h,i}^* = - \underbrace{\sum_{i \in \Theta_b} Q_{b,i}^*}_{\text{ST supply}} \equiv Q^*$$

- In data,  $q^* > 0$  on average  $\Rightarrow$  implies pension funds have net positive swap holdings in equilibrium
- $Q^* \approx 0$  historically,  $Q^* > 0$  in recent years ► ST Eqm

► Frictionless Eqm

► Liq Prem

1. UK Inflation Swap Market

2. Model

3. Identification

4. Estimates and Applications

4.1 Impulse Responses

4.2 Historical Decomposition

# Identification

Problem: observed prices  $(p, P)$  can move either because of **shocks to fundamentals** or **shocks to liquidity frictions**

- Fundamental shock: shock to both  $\pi^e, \Pi^e$
- Liquidity friction (= demand/supply shocks) from each sectors
- Vector of shocks:  $\epsilon = (\epsilon_h, \epsilon_f, \epsilon_b, \epsilon_\pi)'$
- Observables:  $Y = (q, p, Q, P)'$

Then the system:  $Y = \Psi\epsilon \Rightarrow$  Identification: finding elements of  $\Psi_{4 \times 4}$

Three identification strategies:

1. Heterogeneity in reactivity
2. Cross-sectional granularity
3. Heteroskedasticity across time

# Identification

Problem: observed prices  $(p, P)$  can move either because of **shocks to fundamentals** or **shocks to liquidity frictions**

- Fundamental shock: shock to both  $\pi^e, \Pi^e$
- Liquidity friction (= demand/supply shocks) from each sectors
- Vector of shocks:  $\epsilon = (\epsilon_h, \epsilon_f, \epsilon_b, \epsilon_\pi)'$
- Observables:  $Y = (q, p, Q, P)'$

Then the system:  $Y = \Psi\epsilon \Rightarrow$  Identification: finding elements of  $\Psi_{4 \times 4}$

## Three identification strategies:

1. Heterogeneity in reactivity
2. Cross-sectional granularity
3. Heteroskedasticity across time

# 1. Heterogeneity in Reactivity

Assumption: **reactivity** in response to news about fundamentals

Hedge funds > Dealer banks > Pension funds

(Assumption: More attentive/informative  $\Rightarrow$  larger response)

- Dealers (supply) > pension funds (demand) in the LT market
  - Dealers see and trade in both LT, ST markets
  - Likely to have more precise information about inflation
- Hedge funds (demand) > dealer banks (supply) in the ST market
  - Hedge funds are assumed to be more informed traders

$\therefore$  Can impose **sign restrictions** on responses of prices and quantities  
 $\Rightarrow$  Identify elements of  $\Psi$  up to their sign

► Sign Restrictions

## 2. Cross-Sectional Granularity

Assumption: since some institutions in the sample are **larger**, **individual shocks to their demand** function will drive the aggregate demand in the market

- Data has transaction-level information on  $q_{i,t}$ ,  $a_{i,t}$  for each institution  $i$  in each sector  $\Rightarrow$  can identify large institutions

Example: pension fund sector

- Asset demand system can be equivalently expressed as a panel factor model
- Extract idiosyncratic component of each pension fund's demand
- Construct an **instrument for sector-specific shocks** to pension fund demand [▶ Details](#)
- Repeat for hedge funds, dealer banks  $\rightarrow$  one instrument for each sector-specific shock
- There are four potential shocks  $\rightarrow$  fundamental shock appears as a residual

### 3. Heteroskedasticity Across Time

---

Assumption: market participants learn the most about fundamental inflation around data release dates

- Actual fundamentals are revealed on release dates
- Data: 48 dates where monthly inflation data was released + date of announcement of energy cap
  - ⇒ **Spikes in trading activity**, with large price adjustments
- This spike in variance is likely driven by the emergence of **fundamental shocks on these dates**
- Enables identification of fundamental shocks

1. UK Inflation Swap Market

2. Model

3. Identification

4. Estimates and Applications

4.1 Impulse Responses

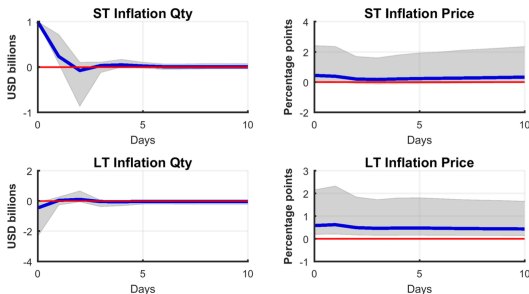
4.2 Historical Decomposition



# Response to a Fundamental Shock ( $\pi^e \uparrow$ , $\Pi^e \uparrow$ )

**Figure 12** ESTIMATED IMPULSE RESPONSE FUNCTIONS TO A FUNDAMENTAL SHOCK

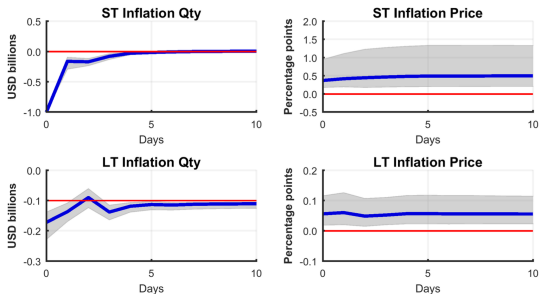
(i) With timing restriction strategy



Median estimate, 68% confidence bands

- Scaled:  $\Delta$  aggr. positions of all ST client institutions = \$ 1 bn [► Detail](#)
- Take 2-3 days for prices/quantities to adjust, then persist over time
- Fundamental shocks permanent & market incorporates new information quickly

# Response to a Liquidity Shock ( $\epsilon_{b,t} \uparrow$ )



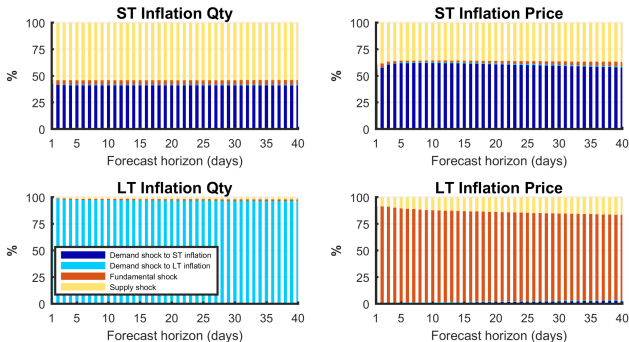
- Corresponds to leftward shift in supply curve in both markets
- Quantity  $\downarrow$ , prices  $\uparrow$  on impact

►  $\epsilon_{f,t}$

►  $\epsilon_{h,t}$

# Variance Decomposition

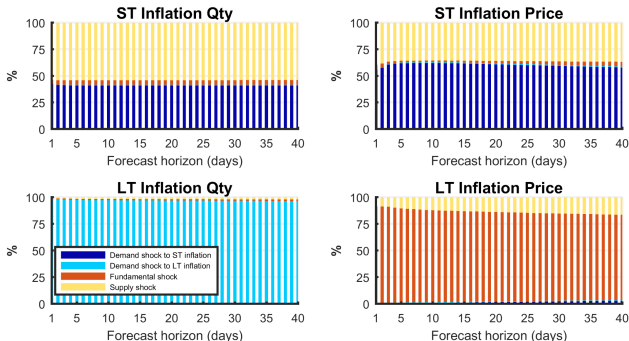
**Figure 15** FORECAST ERROR VARIANCE DECOMPOSITION



- ST: **supply** shocks (dealer banks), **demand** shocks (hedge funds)
- LT: Quantities → **demand** shocks (pension funds)  
Prices → **fundamental** shock ▶ Slopes

# Variance Decomposition

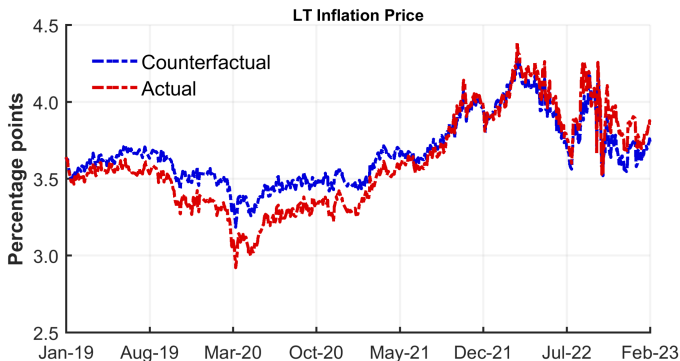
Figure 15 FORECAST ERROR VARIANCE DECOMPOSITION



- Changes in **LT inflation swap rates** are a more reliable indicator of expected inflation, since they mainly reflect fundamental shocks

# Historical Decomposition

**Figure 20** FUNDAMENTAL EXPECTED INFLATION



- Counterfactual: remove all supply/demand shocks [▶ ST](#)
- Possibility that the conventional (actual) measure overstates expected inflation

# Conclusion

- Data → Segmentation in the UK inflation swap market
- Model of the inflation swap market
  - Pension funds, hedge funds, dealer banks choose portfolios
  - Decomposition of swap demand: expected inflation, compensation for risk, liquidity frictions
- Market responds swiftly to new information
- LT swap prices mainly driven by shocks to fundamental expected inflation, while ST price volatility driven by liquidity shocks



Figure 1. Prices of Inflation Swaps in the Recent Past

United States



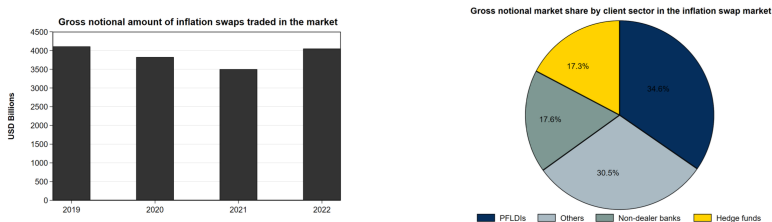
United Kingdom



Zero-coupon swap breakeven rates. Source: Bloomberg

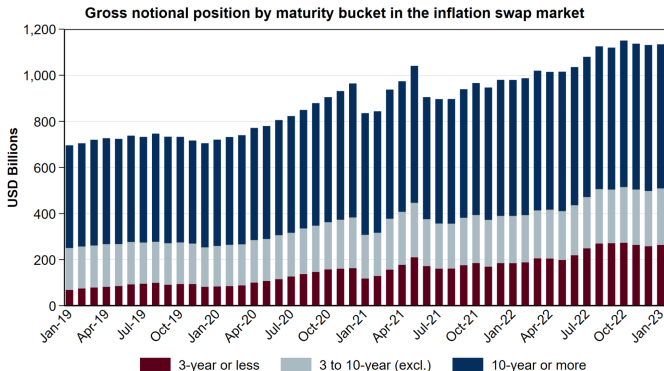


**Figure 2** GROSS NOTIONAL AMOUNTS: AVERAGE AND COMPOSITIONS

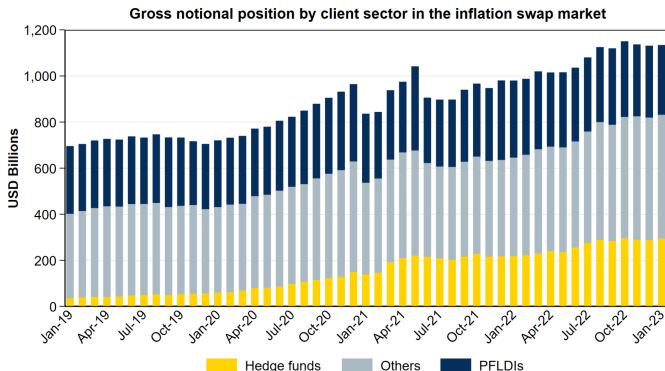


# Market Size and Main Participants

**Figure 3** THE QUANTITIES AND THE INSTITUTIONS BEHIND THE PRICES

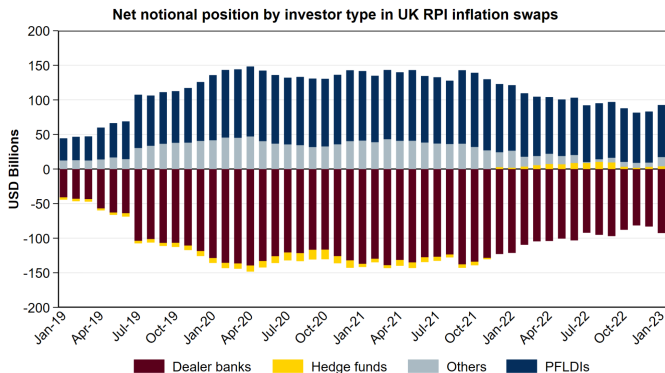


# Market Size and Main Participants



NOTE: Gross notional positions outstanding that are traded in the dealer-client segment of the market across all price indices. SOURCE: DTCC Trade Repository OTC interest rate trade state files, from January 2019 to December 2022.

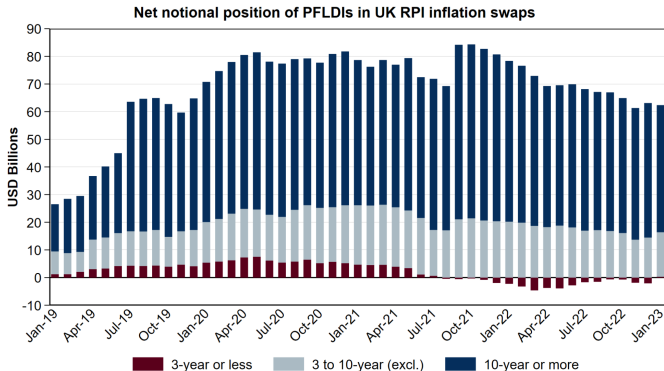
# 1. Dealer banks not neutral [< Back](#)



- Rather, they are net sellers of inflation swaps

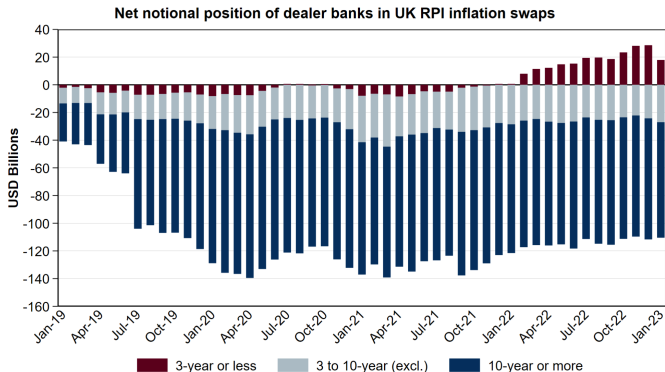
## 2. PFLDIs → Long horizon markets [← Back](#)

**Figure 5** PENSION FUNDS BUY PROTECTION FROM DEALERS IN THE LONG HORIZON MARKET



- Persistently large positive net positions

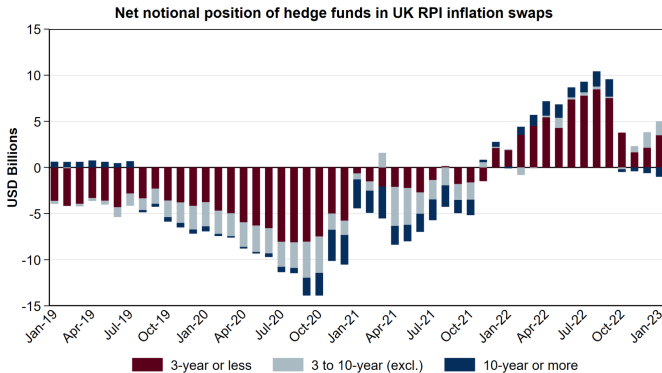
## 2. PFLDIs → Long horizon markets [◀ Back](#)



- Dealer banks hold large amount of inflation risk in LT swaps

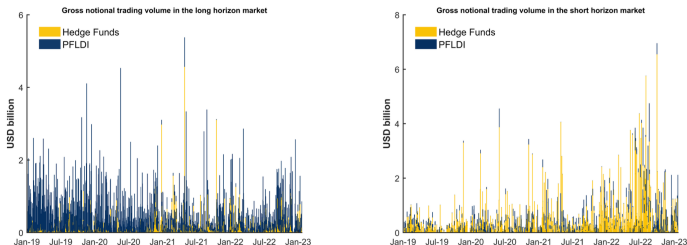
### 3. Hedge funds → Short horizon markets [← Back](#)

**Figure 6** HEDGE FUNDS ARE ACTIVE IN THE SHORT HORIZON MARKET



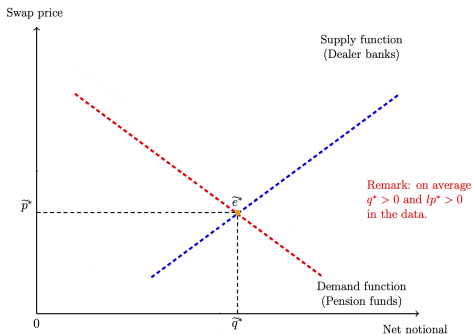
- Switch between net buyers and sellers

**Figure 7** MARKET SEGMENTATION AT HIGHER FREQUENCIES





**Figure 9** THE FRICTIONLESS EQUILIBRIUM



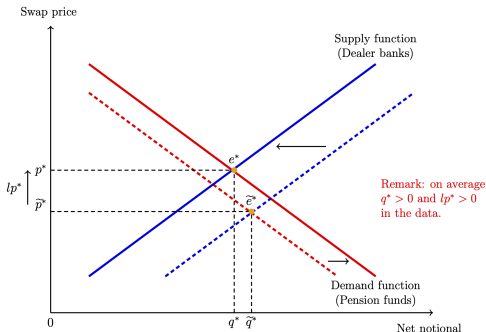
Frictionless eqm: when

- There are complete markets to fully ensure idiosyncratic income risk ( $\sigma_{\pi, y_{f,i}} = \sigma_{\pi, y_{h,i}} = \sigma_{\pi, y_{b,i}} = 0$ )
- Capacity constraints non-binding for any agent in any market (all  $\lambda$ 's = 0)
- Denote frictionless price  $\tilde{p}^*$

Frictionless eqm (in LT market):  $\tilde{e}$

# Liquidity Premium ◀ Market clearing

**Figure 9** THE FRICTIONLESS EQUILIBRIUM



Frictionless eqm:  $\tilde{e}$ , realized eqm:  $e^*$

Shocks to expected inflation or to each sector can shift demand/supply

$\Rightarrow$  Observed prices  $p^* \neq \tilde{p}^*$

**Liquidity premium (LT market):**

$$lp^* = p^* - \tilde{p}^*$$

$$\begin{aligned}
 &= \underbrace{\left( - \frac{\sum_{i \in \Theta_b} \left\{ \sigma_{\pi, y_{b,i}} + \frac{\lambda_{b,i}^L g_{b,i}^L}{\gamma_{b,i}^L} \right\}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} \right)}_{\text{supply friction from dealer banks}} \\
 &+ \underbrace{\left( - \frac{\sum_{i \in \Theta_f} \left\{ \sigma_{\pi, y_{f,i}} + \frac{\lambda_{f,i} g_{f,i}}{\gamma_{f,i}} \right\}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} \right)}_{\text{demand friction from pension funds}}
 \end{aligned}$$

# Frictionless Equilibrium

Lemma 1. **Frictionless price** of a long horizon inflation swap

$$\tilde{p}^* = \underbrace{\left[ \frac{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} \mu_{f,i}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} + \frac{\sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1} \mu_{b,i}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} \right]}_{\text{size-weighted dispersion of beliefs}} \underbrace{\pi^e}_{\text{expected inflation}} - \underbrace{\frac{\theta_d - \tilde{s}^*}{\sigma_d^2} \sigma_{\pi,d}}_{\text{risk premium}}$$

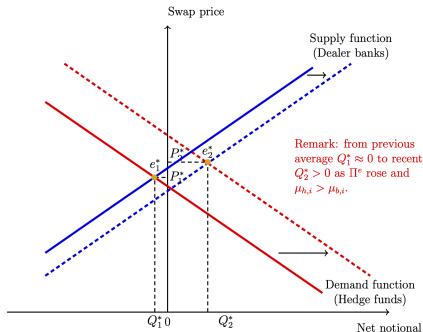
Lemma 2. **Liquidity premium** in the long horizon market

$$lp^* = p^* - \tilde{p}^* = \underbrace{\left( - \frac{\sum_{i \in \Theta_b} \left\{ \sigma_{\pi, y_{b,i}} + \frac{\lambda_{f,i}^L g_{f,i}^L}{\gamma_{b,i}^L} \right\}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} \right)}_{\text{supply friction from dealer banks}} + \underbrace{\left( - \frac{\sum_{i \in \Theta_f} \left\{ \sigma_{\pi, y_{f,i}} + \frac{\lambda_{f,i} g_{f,i}}{\tilde{\gamma}_{f,i}} \right\}}{\sum_{i \in \Theta_f} \tilde{\gamma}_{f,i}^{-1} + \sum_{i \in \Theta_b} \tilde{\gamma}_{b,i}^{-1}} \right)}_{\text{demand friction from pension funds}}$$

# Equilibrium in the Short Horizon

◀ Back

**Figure 10** EQUILIBRIUM IN THE SHORT HORIZON INFLATION SWAP MARKET



- Banks and hedge funds assumed to have similar beliefs on background risk, expected inflation,  $\dots \Rightarrow$  historically  $Q^* \approx 0$
- Recently,  $\Pi^e \uparrow$  ( $\uparrow$  dispersion in beliefs) + demand/supply shocks from e.g. the pandemic

$$Y = \begin{pmatrix} Q \\ P \\ q \\ p \end{pmatrix} = \begin{pmatrix} + & 0 & - & + \\ + & 0 & + & + \\ 0 & + & - & - \\ 0 & + & + & + \end{pmatrix} \begin{pmatrix} \epsilon_h \\ \epsilon_f \\ \epsilon_b \\ \epsilon_\pi \end{pmatrix}$$

- Example: Fundamental shock ( $\pi^e \uparrow$ ,  $\Pi^e \uparrow$ )  
 $\Rightarrow$  LT swap market: Supply shift ( $\downarrow$ )  $>$  demand shift ( $\uparrow$ )  $\therefore p \uparrow, q \downarrow$   
 $\Rightarrow$  ST swap market: Supply shift ( $\downarrow$ )  $<$  demand shift ( $\uparrow$ )  $\therefore P \uparrow, Q \uparrow$

Can rewrite pension fund demand as

$$\frac{q_{f,i,t}}{a_{f,i,t}} = \omega_{f,i} \mathbf{F}_t + \epsilon_{f,i,t} \quad \text{with } \mathbf{F}_t = \begin{pmatrix} \pi_t^e \\ lp_t^* \end{pmatrix}, \quad \omega_{f,i} = \begin{pmatrix} \frac{(\kappa_{f,i}^\mu \mu_{f,t} - \Lambda_t)}{\gamma_{f,i} \sigma_\pi^2 (1 - \rho_{\pi,d}^2)} \\ \frac{(\kappa_{f,i}^{lp} - 1)}{\gamma_{f,i} \sigma_\pi^2 (1 - \rho_{\pi,d}^2)} \end{pmatrix}$$

Estimate the above fixed effects model following Bai (2009)

Construct instrument for sector-wide shocks to pension fund demand  $\epsilon_{f,t}$

$$GIV_{f,t} = \sum_{i \in \Theta_f} a_{f,i,t} \epsilon_{f,i,t}$$

(Can check that this satisfies relevant conditions for valid instrument)

Repeat the process for hedge funds, dealers  $\rightarrow GIV_{h,t}, GIV_{b,t}$

Dynamic model to allow for gradual price/quantity adjustments

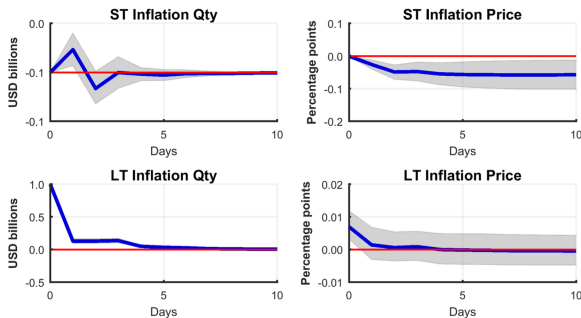
Estimate a Bayesian VAR with diffuse priors,  $L = 3$ :

$$\mathbf{Y}_t = \mathbf{c} + \sum_{l=1}^L \boldsymbol{\Phi}_l \mathbf{Y}_{t-l} + \mathbf{u}_t \quad \text{and} \quad \mathbf{u}_t = \boldsymbol{\Psi} \epsilon_t$$

- $\mathbf{c}$ : a deterministic constant
- Implementing identification strategies
  1. Follow Arias, Rubio-Ramirez, Waggoner (2018) Sign restrictions on  $\boldsymbol{\Psi}$  give set identification
  2. Follow Stock and Watson(2018): Use the GIV as proxy instrumental variables
  3. Pick the shock with properties that are closest to a fundamental shock

# Response to a Liquidity Shock ( $\epsilon_{f,t} \uparrow$ ) [◀ Back](#)

(ii) PFLDI Demand Shock ( $\epsilon_{f,t}$ )



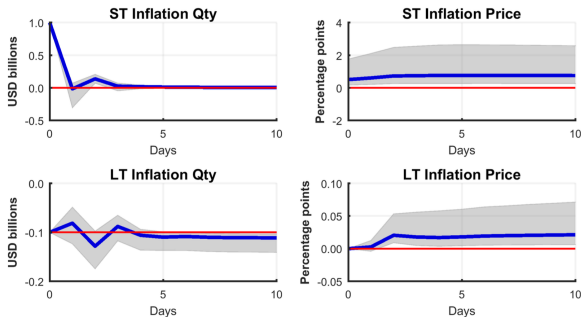
- Scaled  $\rightarrow$   $\Delta$  aggr. positions of all LT client institutions = \$ 1 bn
- Rightward shift of pension funds' swap demand curve
- LT market responses standard: Quantity  $\uparrow$ , prices  $\uparrow$



# Response to a Liquidity Shock ( $\epsilon_{h,t}$ )

[◀ Back](#)

(iii) Hedge Fund Demand Shock ( $\epsilon_{h,t}$ )

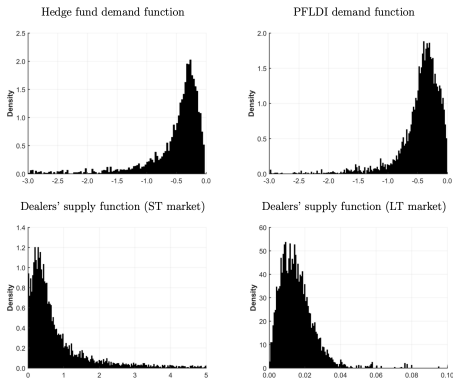


- Rightward shift of hedge funds' swap demand curve
- ST market responses standard: Quantity  $\uparrow$ , prices  $\uparrow$

# Estimated Slopes of Supply/Demand

[◀ Back](#)

Figure 14 DISTRIBUTION OF ESTIMATED SLOPES OF MARKET DEMAND AND SUPPLY



NOTE: Plotted slopes shown are truncated at the given axis values above.

- When a shock hits, compute slope of demand function locally by computing  $d(\text{price})/d(\text{quantity})$ , where  $d(\text{price})$ ,  $d(\text{quantity})$  are responses on impact
- LT supply slope  $\approx 0$ , dealers have almost full bargaining power

# Historical Decomposition (Short Horizon)

[◀ Back](#)